12.16 (a) From the Proca Lagrangian, Eq. (12.91), the canonical stress tensor $T^{\alpha\beta}$ is

$$T^{\alpha\beta} = \frac{\partial \mathcal{L}_{free}}{\partial(\partial_\alpha A^\lambda)} \partial^\beta A^\lambda - g^{\alpha\beta} \mathcal{L}_{free}$$

$$= -\frac{1}{4\pi} g^{\alpha\mu} F_{\mu\lambda} \partial^\beta A^\lambda + \frac{1}{16\pi} g^{\alpha\beta} F_{\lambda\nu} F^{\lambda\nu} - \frac{\mu^2}{8\pi} g^{\alpha\beta} A_\lambda A^\lambda .$$

Since $\partial^\beta A^\lambda = -F^{\lambda\beta} + \partial^\lambda A^\beta$, we have

$$T^{\alpha\beta} = \frac{1}{4\pi} g^{\alpha\mu} F_{\mu\lambda} F^{\lambda\beta} + \frac{1}{16\pi} g^{\alpha\beta} F_{\lambda\nu} F^{\lambda\nu} - \frac{\mu^2}{8\pi} g^{\alpha\beta} A_\lambda A^\lambda - \frac{1}{4\pi} g^{\alpha\mu} F_{\mu\lambda} \partial^\lambda A^\beta$$

In the last term, take into account of the Proca equations of motion for a free particle, $\partial^\beta F_{\beta\alpha} + \mu^2 A_\alpha = 0$, it can be written as

$$-\frac{1}{4\pi} g^{\alpha\mu} F_{\mu\lambda} \partial^\lambda A^\beta = \frac{1}{4\pi} g^{\alpha\mu} F_{\lambda\mu} \partial^\lambda A^\beta = \frac{1}{4\pi} g^{\alpha\mu} \left(F_{\lambda\mu} \partial^\lambda A^\beta + A^\beta \partial^\lambda F_{\lambda\mu} + \mu^2 A_\mu A^\beta \right)$$

$$= \frac{1}{4\pi} g^{\alpha\mu} \partial^\lambda \left(F_{\lambda\mu} A^\beta \right) + \frac{1}{4\pi} \mu^2 A^\alpha A^\beta .$$

Following the same operation as in Section 12.10.B, we can drop the total derivative term, and the symmetric stress tensor becomes

$$\Theta^{\alpha\beta} = \frac{1}{4\pi} g^{\alpha\mu} F_{\mu\lambda} F^{\lambda\beta} + \frac{1}{16\pi} g^{\alpha\beta} F_{\lambda\nu} F^{\lambda\nu} - \frac{\mu^2}{8\pi} g^{\alpha\beta} A_\lambda A^\lambda + \frac{1}{4\pi} \mu^2 A^\alpha A^\beta$$

$$= \frac{1}{4\pi} \left[g^{\alpha\gamma} F_{\gamma\lambda} F^{\lambda\beta} + \frac{1}{4} g^{\alpha\beta} F_{\lambda\nu} F^{\lambda\nu} + \mu^2 \left(A^\alpha A^\beta - \frac{1}{2} g^{\alpha\beta} A_\lambda A^\lambda \right) \right] .$$

(b) From part (a),

$$\partial_\alpha \Theta^{\alpha\beta} = \frac{1}{4\pi} \left[\partial^\alpha (F_{\alpha\lambda} F^{\lambda\beta}) + \frac{1}{4} \partial^\beta (F_{\lambda\nu} F^{\lambda\nu}) + \mu^2 \partial_\alpha (A^\alpha A^\beta) - \frac{\mu^2}{2} \partial^\beta (A_\lambda A^\lambda) \right]$$

$$= \frac{1}{4\pi} \left[(\partial^\alpha F_{\alpha\lambda}) F^{\lambda\beta} + F_{\alpha\lambda} \partial^\alpha F^{\lambda\beta} + \frac{1}{2} F_{\lambda\nu} \partial^\beta F^{\lambda\nu} \right.$$

$$\left. + \mu^2 (\partial_\alpha A^\alpha) A^\beta + \mu^2 A^\alpha (\partial_\alpha A^\beta) - \mu^2 A_\lambda \partial^\beta A^\lambda \right]$$

Using Proca equations of motion, we have

$$\partial_\alpha \Theta^{\alpha\beta} = \frac{1}{4\pi} \left[\left(\frac{4\pi}{c} J_\lambda - \mu^2 A_\lambda \right) F^{\lambda\beta} + \underline{F_{\alpha\lambda} \partial^\alpha F^{\lambda\beta} + \frac{1}{2} F_{\lambda\nu} \partial^\beta F^{\lambda\nu}} \right.$$

$$\left. + \mu^2 (\partial_\alpha A^\alpha) A^\beta + \mu^2 A^\alpha (\partial_\alpha A^\beta) - \mu^2 A_\lambda \partial^\beta A^\lambda \right]$$

The underscored terms give a result of zero, as shown in Section 12.10.C. Then,

$$\partial_\alpha \Theta^{\alpha\beta} = \frac{1}{c} J_\lambda F^{\lambda\beta} + \frac{1}{4\pi}\Big[-\mu^2 A_\lambda (\partial^\lambda A^\beta - \partial^\beta A^\lambda) + \mu^2 (\partial_\alpha A^\alpha) A^\beta$$
$$+ \mu^2 A^\alpha (\partial_\alpha A^\beta) - \mu^2 A_\lambda (\partial^\beta A^\lambda) \Big]$$
$$= \frac{1}{c} J_\lambda F^{\lambda\beta} + \frac{1}{4\pi}\mu^2 (\partial_\alpha A^\alpha) A^\beta .$$

From the equations of motion, we know

$$\partial_\nu \partial_\mu F^{\mu\nu} + \mu^2 \partial_\nu A^\nu = \frac{4\pi}{c} \partial_\nu J^\nu , \quad \text{or} \quad \partial_\nu J^\nu = \frac{c}{4\pi} \mu^2 \partial_\nu A^\nu .$$

For charge conservation, we must have $\partial_\nu A^\nu = 0$. which finally leads to

$$\partial_\alpha \Theta^{\alpha\beta} = \frac{1}{c} J_\lambda F^{\lambda\beta} .$$

(c) For $\alpha = \beta = 0$,

$$\Theta^{00} = \frac{1}{4\pi}\Big[g^{0\gamma} F_{\gamma\lambda} F^{\lambda 0} + \frac{1}{4} F_{\lambda\nu} F^{\lambda\nu} + \mu^2 \big(A^0 A^0 - \frac{1}{2} A_\lambda A^\lambda\big)\Big]$$

The first part is identical to Θ^{00} of an electromagnetic field, and notice that

$A_\lambda A^\lambda = A^0 A^0 - \vec{A}\cdot\vec{A}$, then

$$\Theta^{00} = \frac{1}{8\pi}(E^2 + B^2) + \frac{1}{4\pi}\Big[A^0 A^0 - \frac{1}{2}(A^0 A^0 - \vec{A}\cdot\vec{A})\Big] = \frac{1}{8\pi}\Big[E^2 + B^2 + \mu^2 (A^0 A^0 - \vec{A}\cdot\vec{A})\Big] .$$

Similarly, for $\alpha = i$, $\beta = 0$.

$$A^i A^0 - \frac{1}{2} g^{i0} A_\lambda A^\lambda = A^i A^0 .$$

Then, $\Theta^{i0} = \dfrac{1}{4\pi}\Big[(\vec{E}\times\vec{B})_i + \mu^2 A^i A^0 \Big]$.